GOAL SETTING and ALTERNATE CAREER PLAN WORKSHEET (10%)

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CO-OP TERM GOALS SETTING WORKSHEET (5%)

Where am I now?
(Where are you in terms of education, skills and experience as it pertains to your program or field of study? What is the larger scope of where you are? Be specific.)

In a bizarre world where life can be settled, the dreams can be achieved, the skills can be enhanced, and the experience would be great. A field which demands a lot of hard work, smartness, focus, and struggle, although with an experience of real world how to develop a software, especially from scratch.

End Point (Co-op) – Where do I want to be at the beginning of my work term?
(Note: This can be the path toward your ultimate career goal. Where do you want to be at the beginning of your co-op term? Here you want to include the type of role and type of company you wish to work for. What requirements do you need? What skills and experience are these positions asking for? What do you hope to gain from this experience?)

I do not have much idea about the organizations in Canada, but I would like to begin my journey either in a startup company where I would be getting more opportunities to learn but struggle, or in a well-set organization where I would be able to deal with various tech-leads and understand the professional world more precisely. I wish to start as a junior developer in order to get trained, understand, and thoroughly apply the concepts I have learned so far while also getting an exposure of gaining knowledge and experience of other legends in the field to grow in a better way. It would require me not only the technical skills but also the basic skillset such as communication, gestures, professionalism, behaviour, and teamwork. While my past experience would help in leading my way on performing technical tasks, the Canadian culture and its environment will bring more clarification on the workstyle and growth opportunities.

How can I get there (identify Goals with 5 points i.e. S.M.A.R.T on how you will achieve them)
Instructions: Each of your goals should be a S.M.A.R.T goal. Please refer to the url provided in Class 4: What are SMART goals?

1. **GOAL** Developing General Purpose Artificial Intelligence that can perform beyond human capabilities and showcasing God-level power while also some way to control it to not push beyond its limits or leadings its way towards either vast disasters or global extension.

a. **Specific:** Built a general-purpose AI using either Python or in a custom environment which includes a customized OS, a custom programming language, and a custom hardware interface and drivers for more efficient working.

b. **Measurable:** Two ways, either develop a general-purpose model from scratch which would take a lot of time and efforts to work correctly, or simply develop special purpose AI with a note in the mind that how it would be integrated with other AI models to build a general-purpose AI. So basically, a collection of special purpose AI.

c. **Attainable:** Focusing on major special purpose AI models and understanding and building them on my own. The initial models in my opinion would be on image-processing and speech-recognition. In parallel, learning computer vision and building natural language processing algorithms would help my further in evolving my knowledge on how to make these AI in order to gain wisdom.

d. **Relevant:** This intelligent system would be capable of duplicating my existence by simply performing the tasks that I should, but the machine would on its own and in that way, I would be able to grow exponentially in my life and would be able to perform several activities at once in the entire globe.

e. Time-Bound: Since I do not know how to make even a simple AI, it is hard to predict how long would I be taking to make such a stable and controlled system. However, I would target to develop one by the end of 2030 or 2035. My initial goals are to understand this sector, which I suppose would be done by 2025, and then, I can join initiate working on this project by developing various special-purpose AI by 2028.

2. GOAL Game (AI) Developer.

a. Specific: AI is used in many sectors, one of those and widely used sector is gaming. Being an AI developer in terms of dynamic (AI based) animation and voice reaction sounds interesting to me. Apart from that, AI is useful in pedestrian action and fidelity improvisation, which would save a lot of time and efforts.

b. **Measurable:** I would love to join Rockstars Games for contributing to their new projects. In order to do that, I would need to develop contacts with various game developers, especially those who are/were working in Rockstar Games which will increase my chance in getting selected.

c. **Attainable:** Establishing network with various developers, especially with Rockstar Games developers to learn the huddles they got and how to deal with it, while also learning the technicality on how to develop games.

d. **Relevant:** I love to develop games. I also have developed various mods for the game Grand Theft Auto San Andreas and Grand Theft Auto Vice City. The most important reason would be I am crazy about games while also I understand how the user feels about the game because I was one of them at one time.

e. **Time-Bound:** I expect this to be done by the end of January or February because developing a game does require quite a beefy hardware so that I could render it at its high potential while also dealing with multiple factors applied in a game.

3. GOAL Music (AI) Engineer. Apply AI in music.

a. **Specific:** Since I am a music-lover, I would like to join the organization *ORG24* which is a music studio in the mobile platform and join their AI team which is working on implementing music with the help of human's thinking, behaviour, feelings, and history (music likings and hates), and contribute in it.

b. **Measurable:** Since I know how to use this software, I can expand my knowledge on experiencing each and every feature it is providing and keep up-to-date with it. Apart from that, I can read articles and research on how AI can contribute to music and how to develop such an AI. Furthermore, I will connect with the developers who are currently contributing to this field and try to catch with them so that it will become easier for me to join and flawlessly contribute.

c. **Attainable:** Firstly, I must start learning variations in music and how a person would expect the music to be. This will better allow me to visualize how people would expect from an AI to provide some music depending on their mood or some other reason. Secondly, building a network with people in this field would push me more towards attaining success in this sector. Lastly, creating a custom project in which I can start developing a prototype on how a system can fetch the human emotions and provide some music onto that.

d. **Relevant:** Not only I love music but also, I am playing various instruments since my childhood. This will be a joyful job and a career with enjoyment becomes a path towards the destiny, the endgame is always happy, and if I would be turning around to see what I have achieved in my life at my old age, I would not be feeling that I made a wrong decision.

e. **Time-Bound:** I would tight myself to begin studying on this sector from today. However, if I fail to achieve any of the above goals, my target would be to study and start diving onto the project and build a prototype within 6 months.

4. GOAL Database Engineer at Database Warehouse.

a. **Specific:** I want to become a Junior Database Engineer at *DataBricks*, an organization which deals with workload distribution based on game traffic to avoid extra computational cost which helps to play games on cloud.

b. **Measurable:** Learning how to collect, organize, store, retrieve, and handle the big data would be by initial step because they are the vitals. Furthermore, I will start commuting with field experts to understand how to work in this sector, determining the professional approaches and concepts, and exploring the AWS server by performing various operations on a big dataset.

c. **Attainable:** Learning various concepts of a database engineer and attaining a certificate on it. I can express my experience on SQL and MongoDB, a relational and non-relational database, which will put some weight on my resume. Apart from that, asking the industry experts about how to progress in this field would add an extra layer of intelligence.

d. **Relevant:** I love dealing and playing with data and as a database engineer, it is important to understand the importance of data and how we can manage it. I have dealt with my data by storing it to various places, creating its replicas, and performing disk-checks to keep it healthy.

e. **Time-Bound:** In my opinion, I can achieve some good results within 150 man-hours.

ALTERNATE CAREER PLAN GOAL SETTING WORKSHEET (5%)

As an AI and ML specialist, you can plan for any of these alternate careers (these are suggestions, you can look into other areas of interest as well): For your alternate career plan (not your first choice, but your second choice, please research these on indeed.ca or Onetonline.org to find the job qualifications and requirements.)

- AI and ML Developer
- AI and ML Technician
- Data Engineer
- AI and ML Analyst
- AI Engineer
- AI Developer
- Software Engineer
- Machine Learning Engineer
- Data Analytics Developer
- Software Tester
- Mobile Application Developer
- Computer Programmer
- Systems Analyst
- Data Analyst
- Data Science Developer
- Database Developer
- Web Application Developer

What is my alternate career choice?
(Choose an alternate career. Be specific.)

I have developed myself to be a flexible person and switching between various technical professions as a developer is common in the software industry. If not as an *AI and ML developer*, I would like to become a *game* 🎮 *developer* developing the business logic in games on how the game's logic has to be performed, a *full stack* 💻 *DotNet developer* who can develop various applications with the help of DotNet Framework, a *flutter* 📱 *developer* to develop cross-platform web and mobile applications, or at a worse case scenario, I would love to be a *database* 🗄️ *engineer* at a data warehouse whose job would be to handle the bulky big data.

Qualifications (Alternate Career) – What are the skills and qualifications required for this alternate career choice?
(Note: How will you plan to ensure that you have these qualifications and skills?)

In my opinion, I would need a great communication, knowledge about various trends in the organizations regarding coding and architectures, blender for animations, MAUI for becoming a *full stack DotNet developer*, Unreal Engine 5 (at least) for becoming a *game developer*, Google Play Store and Apple Store with AWS for becoming *a flutter developer*, and data-handling techniques for handling big data if I become a *database engineer*.

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1. **GOAL** Flutter 📱 Developer.

   a. **Specific:** Develop various applications that are helpful in mobile and web platforms.

   b. **Measurable:** Learning the famous language Dart, developing various small but useful projects in it, deploying them with the help of Google Play Store, Apple Store, and AWS servers, and building a network with flutter developers to improvise my skillset.

   c. **Attainable:** Learning various platforms and gaining insights of Google Play Store, Apple Store, and AWS for improvising the deployment process while developing open-source projects and attracting various developers to improvise me by providing technical feedbacks.

   d. **Relevant:** I was interested in developing various applications but in order to do that I had to write different code in different ways to deploy for various platforms. Flutter does the same while also providing robustness and mobility. This thing blew my mind and since than I decide that I will be developing at least a few applications that would help me a lot in my day-to-day tasks while also as a skill can earn some 💵 greens.

   e. **Time-Based:** In order to become a flutter developer and develop various applications, I suspect by the end of September 2024, especially with the help of summer break, my flexibility for learning and deploying various projects will increase and would be able to learn and develop then after. However, this seems a bit far, so my current target is to learn it in-between the holidays such as weekends where I can concentrate in learning the Dart language and parallelly developing some ideas on how to apply this knowledge in order to convert it into wisdom.

2. **GOAL** I would be glad to be a C# (WPF) developer since it was one of the works, I was doing for past one year.

   a. Specific: Developing desktop applications has seen a lean for last couple of years. Unfortunately, most people use laptops or desktops for their daily routine. Moreover, they have started using systems that can provide performance and what I have seen from my experience is not every application is optimized to utilize the resources it has and execute more productively while also saving the resources when performing light-weight tasks. Basically, smart move for an optimised and smart application, is what lacking nowadays in various applications.

   b. Measurable: Two ways, either I can join as an old-fashioned framework which is *WPF*, or I adapt the new framework, which is *MAUI*, that not only helps in developing the desktop-based applications but also helps in deploying the same application in various platforms and on web as well.

   c. Attainable: Focusing on any of the technologies would help me to grow in this field while deploying applications optimized for the new and hybrid architectures. My target would be to begin

with MAUI framework, and then, leading my way towards learning web development in C# to be qualified as a full-stack developer and gradually learning AWS to deploy and maintain such softwares.

d. Relevant: I have a strong grip in C# and WPF which not only would generate an ample amount of revenue, but also would be more satisfying and enjoying while working on something in which I have a significant experience. Apart from that, I know some seniors in this field who will help in growing in this sector.

e. Time-Bound: As soon as I perceive my Co-Up, I would jump into this sector, which is 2025. I would gradually like to contribute to this sector by developing applications for companies like AMD and NVIDIA because since AMD's UI is great and the application runs flawlessly, NVIDIA, on the other hand, has very simple UI but a strong connection with the corresponding hardware.

3. **GOAL** I would like to develop games and do cinematography using Unreal Engine with the help of C#.

a. Specific: I love playing games and when I was a kid, I love to know why this is happening and how. This allowed me to come in this field and now my goal is to develop and deploy some of the games on my own that would be a masterpiece in the near future.

b. Measurable: I would first start my journey by learning Blender and Unity softwares which will help me to understand how to develop a game, and then I will jump into developing open-world games which will includes various features. Afterwards, I would jump into Unreal Engine and develop graphically attractive and appealing games.

c. Attainable: Commuting with the people having knowledge or experience in game development in order to being my journey and define a pathway to travel, deploying games on Epic Games and Steam to see how people reacts on my development, and improvising with the help of those feedbacks and training.

d. **Relevant:** I love to see the beauty of fidelity rendered by the machine while also love to do cinematography with such game engines and with the help of this I would be able to generate some side-income which will help in contributing more to developing such tools and softwares or might on my leisure.

e. **Time-Bound:** I would like to set my deadline as the end of February because this does take quite a time to develop and deploy a complete game. I would like to deploy my first game on Steam and later to Epic Games using Unreal Engine by the end of September 2024.

4. **GOAL** IoT Developer (Python or C).

a. **Specific:** If none of the above goals are achievable, I can simple become an IoT developer and with the help of Python or C, I can develop an ample number of automotive tools by joining the company *Raspberry Pi* as a Junior IoT Developer and contributing to developing

various such board while working as a free lancer for organizations for setting up IoT devices and deploying for various businesses.

b. **Measurable:** To begin with, I will start learning IoT development using Python and develop some small projects for practice. Furthermore, I will start doing free lancing and deploy some IoT based projects.

c. **Attainable:** Firstly, I will start learning Python that helps in IoT development. Secondly, I will gain a certification course that will help me for employment purpose. Thirdly, I will start free lancing and develop some custom projects for nearby restaurants where I will narrate the idea of how IoT will help in their business and provide them free trails. Lastly, I will build network with IoT developers who will guide me in order to generate some revenue. Finally, I will apply for organizations which are leading in developing IoT based projects.

d. **Relevant:** Nowadays, the world without IoT device would progress slowly. Additionally, I have a hands-on experience in this field. Apart from that, I love working with devices that would generate big data, which leads towards AI development. So, one way or another, by choosing this field, I can return to my field for what I was meant for.

e. **Time-Bound:** I would target myself to start working on this field and ending up being a free-lancing-ready within 4 months of given time.